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PERFECT SCORE OF 150/150 AS A TESTAMENT TO EXECEPTIONAL E-LEARNING METHODS

UPES UNIVERSITY OF TOMORROW

Elements of AIML

Unit 02 – Lecture 01: Propositional Logic Foundations (Entailment)

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February 28, 2026

Topic focus: **Logic for AI** (syntax, semantics, and basic inference)

Repository: `https://github.com/tali7c/Elements-of-AIML`

Sources: Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 7–9 (Logical Agents, First-Order Logic, Inference).;Huth and Ryan, *Logic in Computer Science* (2e), selected sections on propositional and first-order logic.;Genesereth and Nilsson, *Logical Foundations of Artificial Intelligence*, selected chapters.

Quick Links

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Agenda

Overview

Propositional Logic

Limits of propositional logic (preview)

Practice

Summary

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- Convert simple formulas to **CNF** (preview of automated inference).
- Explain (at a high level) why propositional logic is insufficient for many AI knowledge bases.

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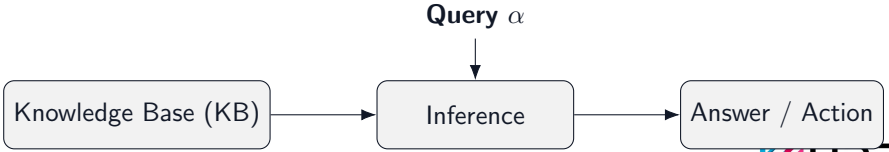
- PL: propositional logic, CNF: conjunctive normal form

Why Logic in AI?

- AI systems often need **explicit knowledge**: facts + rules (a **knowledge base**).



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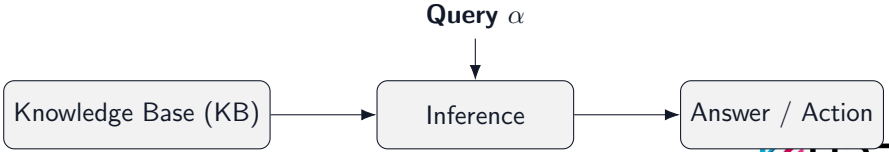

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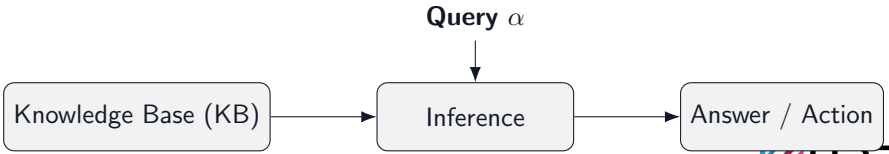

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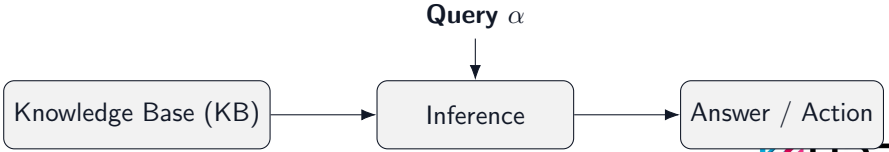

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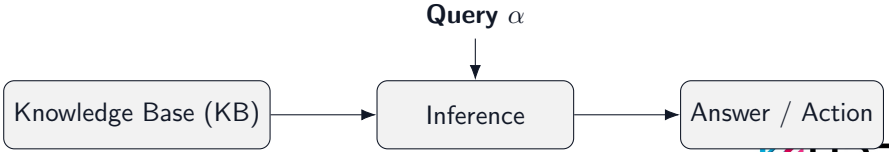

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- In AI we care about **entailment**:

$$KB \models \alpha \quad \text{(in every model where } KB \text{ is true, } \alpha \text{ is true).}$$

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- We build compound sentences using **connectives**:

$$\neg p,\ p \wedge q,\ p \vee q,\ p \rightarrow q,\ p \leftrightarrow q$$

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- They let us check:
  - **Tautology**: always true
  - **Contradiction**: always false
  - **Satisfiable**: true in at least one assignment (has a model)

Implication ($p \to q$) Confusion Fix

$p \to q$ is **false only when** p is true and q is false.

p	q	$p \to q$
T	T	T
T	F	F
F	T	T
F	F	T

Key equivalence:

$$(p \to q) \equiv (\neg p \vee q)$$

Worked Truth Table (Example)

Example formula: $(p \wedge q) \vee \neg p$

p	q	$p \wedge q$	$\neg p$	$(p \wedge q) \vee \neg p$
T	T	T	F	T
T	F	F	F	F
F	T	F	T	T
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Equivalence Laws (Selected)

- De Morgan:

$$\neg(p \wedge q) \equiv (\neg p \vee \neg q), \quad \neg(p \vee q) \equiv (\neg p \wedge \neg q)$$

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- Double negation: $\neg(\neg p) \equiv p$

Normal Forms: CNF vs DNF

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- Many inference procedures (e.g., **resolution**) expect CNF.

CNF Conversion (Mini Example)

Convert: $(p \to q) \land p$

- Eliminate implication: $(\neg p \lor q) \land p$

This is the standard form used for many automated checks.

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- Already in CNF:

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Inference: Entailment vs Derivation

- **Semantic entailment**: $KB \models \alpha$ (meaning-based, via models).

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- Example (Modus Ponens):

$$(p \rightarrow q),\ p \ \vdash \ q$$

Checkpoint #1 (30 seconds)

Given: $KB = \{(p \to q), (q \to r), p\}$

Question: Does $KB \models r$? Briefly justify (1–2 lines).

Why Propositional Logic Becomes Insufficient

Propositional logic is great for **small** worlds, but it struggles when we need:

- **Many objects**: one symbol per object-fact becomes too verbose

Next lecture: **predicate (first-order) logic** solves these with objects, predicates, and quantifiers.

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- **Relations**: "x is parent of y" cannot be expressed naturally
- **Quantification**: "all" / "some" statements cannot be expressed compactly

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Preview: What Predicate Logic Adds (One Slide)

- Predicates: $Student(x)$, $Enrolled(x, y)$

We will do full syntax, semantics, and translation in **Lecture 02**.

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Preview: What Predicate Logic Adds (One Slide)

- Predicates: $Student(x)$, $Enrolled(x, y)$
- Quantifiers: $\forall$ (for all), $\exists$ (there exists)
- Example rule (compact):

$$\forall x\ (Human(x) \rightarrow Mortal(x))$$

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Mini Demo (Propositional Logic by Brute Force)

We can check entailment by enumerating truth assignments (small KB only).

Run:

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python demo/prop_logic_demo.py
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What you'll see:

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- Whether $KB \models \alpha$ holds (and a counterexample model if not)

Practice Problems (In-Class)

1. Build a truth table and decide if the formula is a tautology:

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3. Translate and simplify: "If it is not raining then the ground is not wet."

Summary (Takeaways)

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- **Predicate logic**: adds **objects + relations + quantifiers** $\Rightarrow$ much more expressive.
- Inference is about answering: $KB \models \alpha$ (what follows from what we know).

Exit Questions

1. What does $KB \models \alpha$ mean (in one sentence)?

Sources: Russell and Norvig, *Artificial Intelligence: A Modern Approach* (4e), Ch. 7–9 (Logical Agents, First-Order Logic, Inference).;Huth and Ryan, *Logic in Computer Science* (2e), selected sections on propositional and first-order logic.;Genesereth and Nilsson, *Logical Foundations of Artificial Intelligence*, selected chapters.

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4. Give one limitation of propositional logic that motivates predicate logic.

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Thank You